

## 9 Counting And Probability

### 9.1 Introduction

#### Example 9.1.12

- a) List the eight possibilities:  $\{BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG\}$   
b)  $E_1 = \{BBG, BGB, GBB\}$ . Chance of one being girl:  $\frac{3}{8}$ .  
 $E_2 = \{BGG, GBG, GGB, GGG\}$ . Chance of being *at least* two girls:  $\frac{1}{2}$

#### Example 9.1.13

The monty hall problem.. Woo.

#### Theorem 9.1.1

$m, n \in \mathbf{Z} \wedge m \leq n \Rightarrow$  There are  $n - m + 1$  integers from  $m$  to  $n$  inclusive.

#### Exercise 9.2.5

Creating Trees.